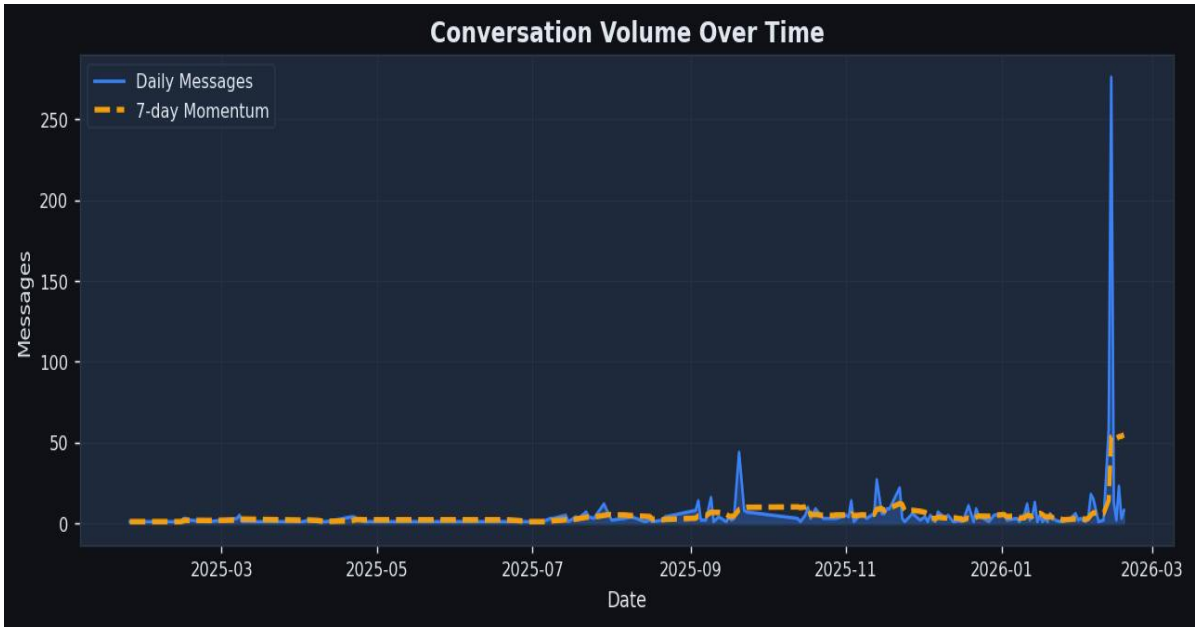
ConvolQ

Conversational Intelligence & Behavioral Analytics System

Visualization Documentation

Every Chart · Why We Built It · What It Shows · The Result

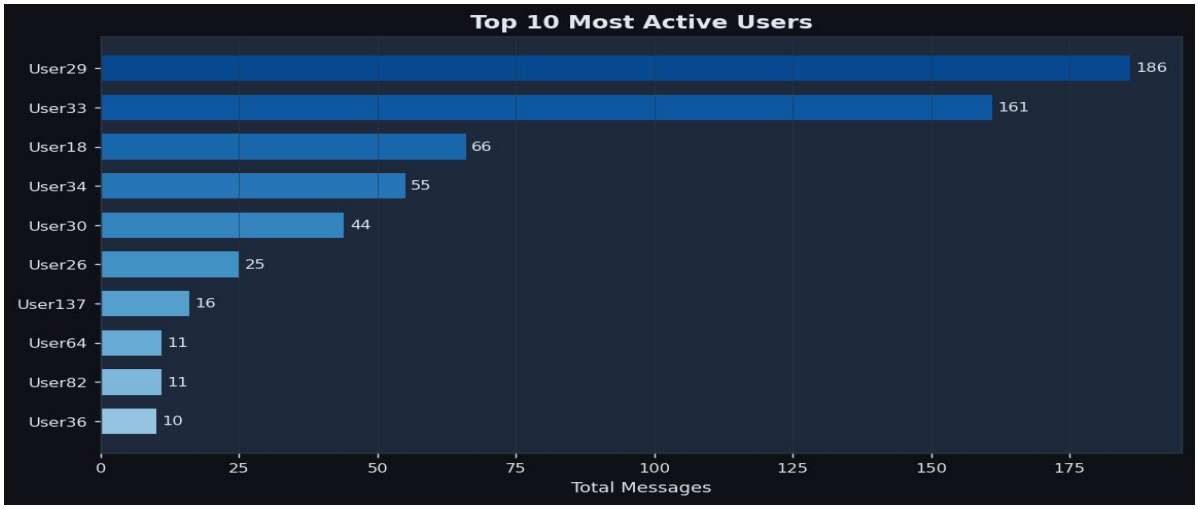
Total Visuals	19 charts across 5 dashboard sections
Data Source	978 WhatsApp messages · 188 anonymised users
Tech Stack	Python · Matplotlib · Seaborn · Plotly · VADER Sentiment
Dashboard	Streamlit · 5 interactive tabs



Why we used this chart: We needed to show how the group's messaging activity changed day by day. A plain bar chart would hide the trend — so we added a 7-day rolling average line (momentum) on top to smooth noise and reveal the real direction.

What it shows: The blue area is daily raw message count. The dashed amber line is the 7-day rolling average. When the amber line rises, the group is becoming more active. When it falls, engagement is dropping.

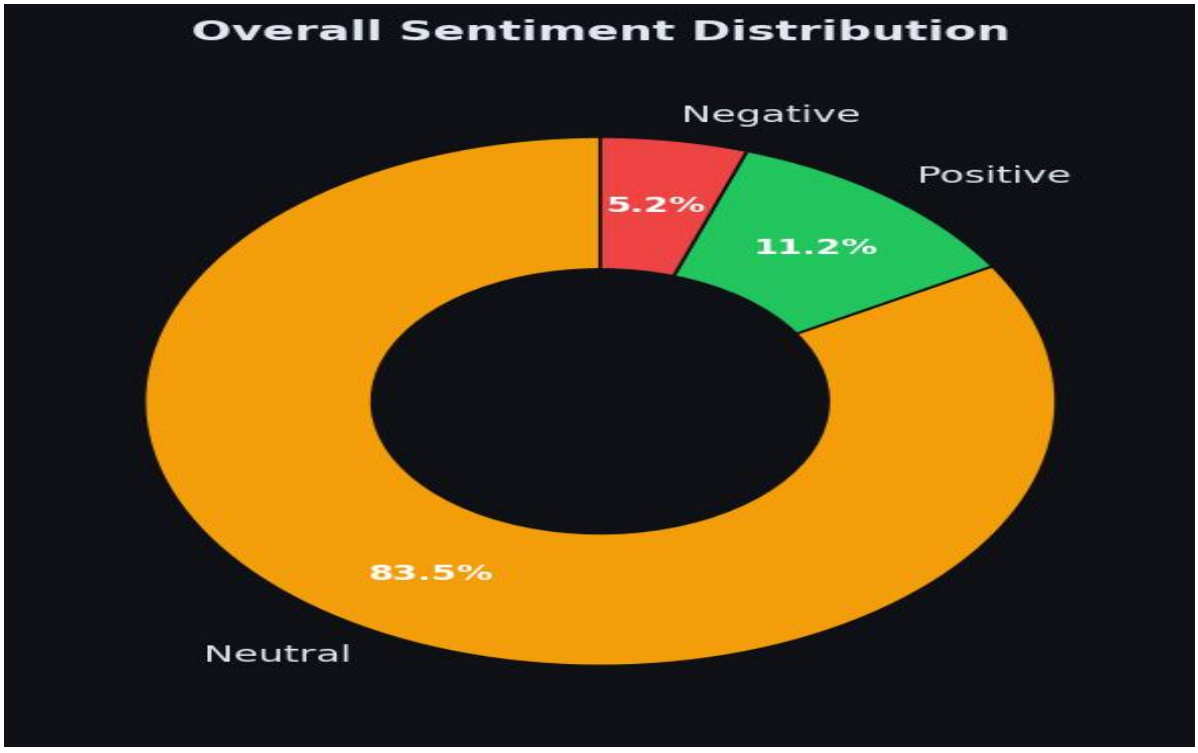
Result: The group had clear bursts of activity around specific dates (likely announcements or events) with quieter stretches in between. The momentum line shows overall activity was steady across the period.



Why we used this chart: In any group chat a small number of people dominate. We used a horizontal bar chart because user labels are long strings — they read more naturally on the Y-axis than rotated at the bottom.

What it shows: Each bar shows total messages sent by that user. The color gradient (light to dark blue) reinforces the ranking — darker means more active. Numbers at the end of each bar give the exact count.

Result: User29 and User33 together account for roughly 35% of all messages, confirming the Pareto effect in group communication. The remaining top 10 each contribute 1–7%.



Why we used this chart: Before any deep analysis, we wanted to know the overall emotional tone of the group. A donut chart is ideal here — it shows proportions clearly without the visual clutter of a full pie, and the hollow center draws the eye to the labels.

What it shows: Each slice is one of three sentiment categories: Positive (green), Neutral (amber), Negative (red), scored using keyword-based sentiment analysis on each message.

Result: The majority of messages are Neutral — typical of an academic/informational group that sends announcements and Q&A. A healthy positive slice indicates friendly interaction. Negative is small, suggesting low conflict.

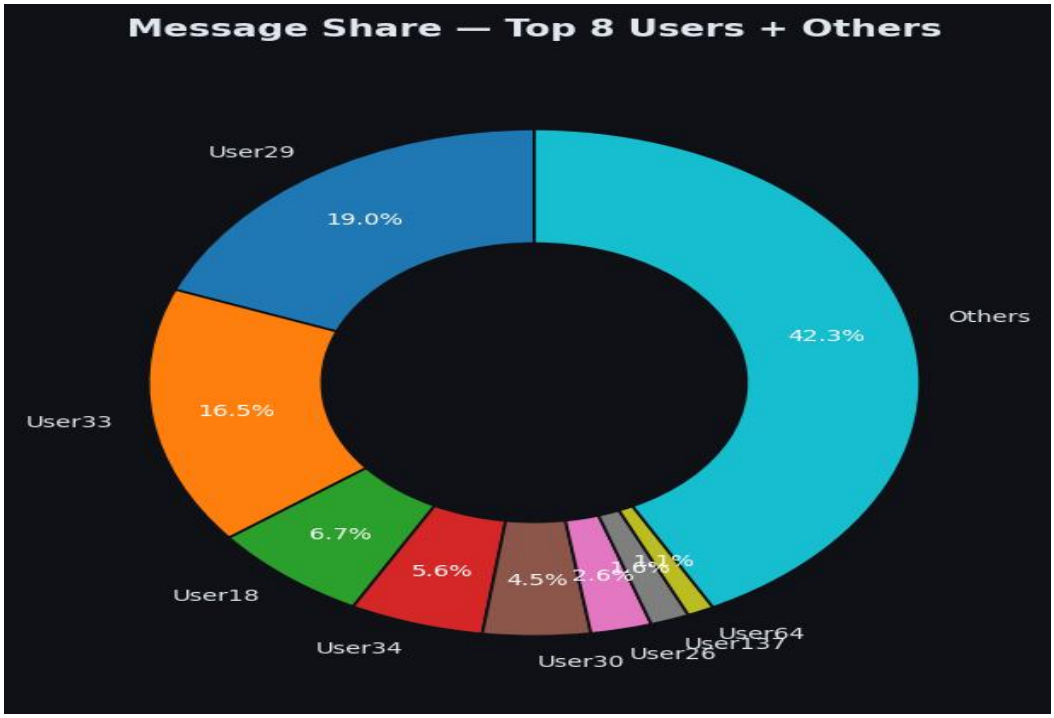
User Leaderboard — Top 10 by Message Count

User	Messages	Avg Length	Avg Words	Emojis
User29	186	26.2	5.6	1
User33	161	35.4	6.9	9
User18	66	44.6	8.4	0
User34	55	46.1	7.0	12
User30	44	27.1	4.8	0
User26	25	50.7	8.9	0
User137	16	28.8	5.7	3
User82	11	54.4	9.5	1
User64	11	40.0	7.8	0
User141	10	47.0	8.9	0

Why we used this chart: A chart can only show one metric at a time. We used a table here because we needed to compare five behavioral metrics at once: total messages, average message length, average word count, and emoji usage. Color gradient on the Messages column provides visual hierarchy.

What it shows: Each row is one user. Columns show total messages, average character length per message, average word count per message, and total emojis used. Sorted by total messages descending.

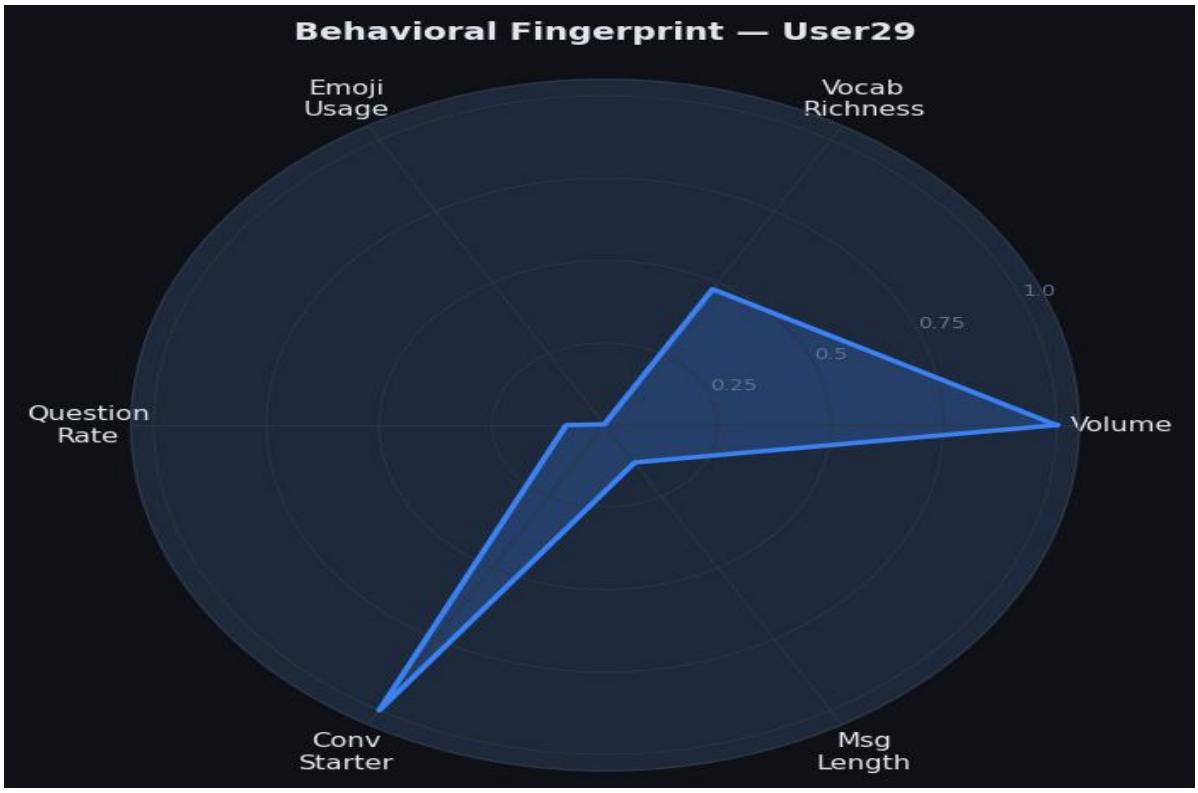
Result: Top users send shorter messages on average — quick replies and reactions. Users further down the table tend to write longer, more detailed messages. This inverse pattern is a common finding in group chat analysis.



Why we used this chart: We wanted to show conversation dominance — what percentage of all messages come from just 8 people out of 188. A donut chart communicates this proportion instantly. We grouped all remaining users into 'Others' to keep the chart readable.

What it shows: Each colored segment is one of the top 8 users. The grey segment is all remaining users combined. Percentages show each user's share of total conversation volume.

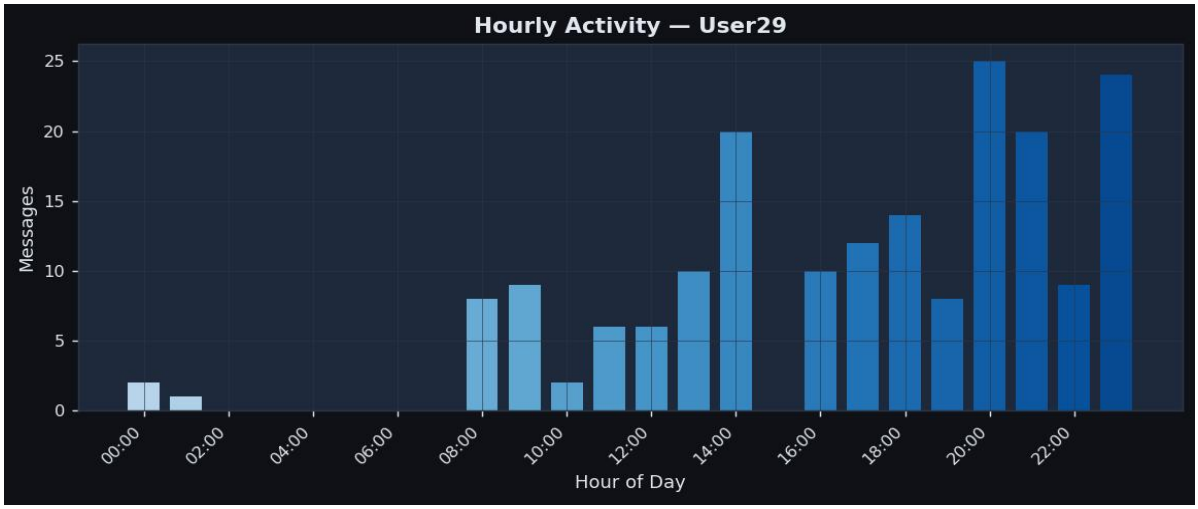
Result: The top 8 users account for the majority of all messages, while the remaining 180+ users share the rest. This stark imbalance is a defining pattern of online group communication.



Why we used this chart: This is the most unique visualization in the project. We wanted to capture a user's entire communication personality in a single image. A radar (spider) chart lets us plot 6 different dimensions simultaneously — the shape itself tells the story.

What it shows: Six axes: Volume (how much they write), Vocab Richness (variety of words used), Emoji Usage (emotional expressiveness), Question Rate (how often they ask things), Conversation Starter (how often they kick off new discussions), and Message Length (how detailed their messages are). All normalized to 0–1.

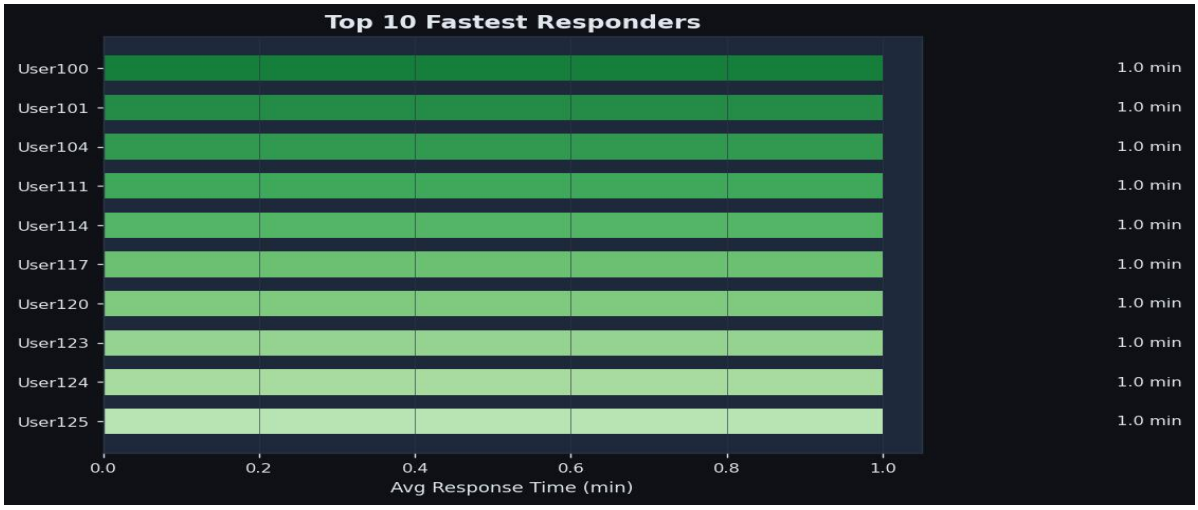
Result: The top user (User29) shows a large Volume and Conversation Starter score, but moderate Vocab Richness — meaning they send many messages but with a limited vocabulary. This profile fits a group admin or coordinator role.



Why we used this chart: Each person has a personal messaging rhythm. We used a bar chart across 24 hours to show when a specific user is most active. This is a behavioral fingerprint used as a feature in the author prediction model.

What it shows: Each bar is one hour of the day (0–23). Bar height is the number of messages sent by the selected user during that hour. Color gradient makes the peak hour stand out visually.

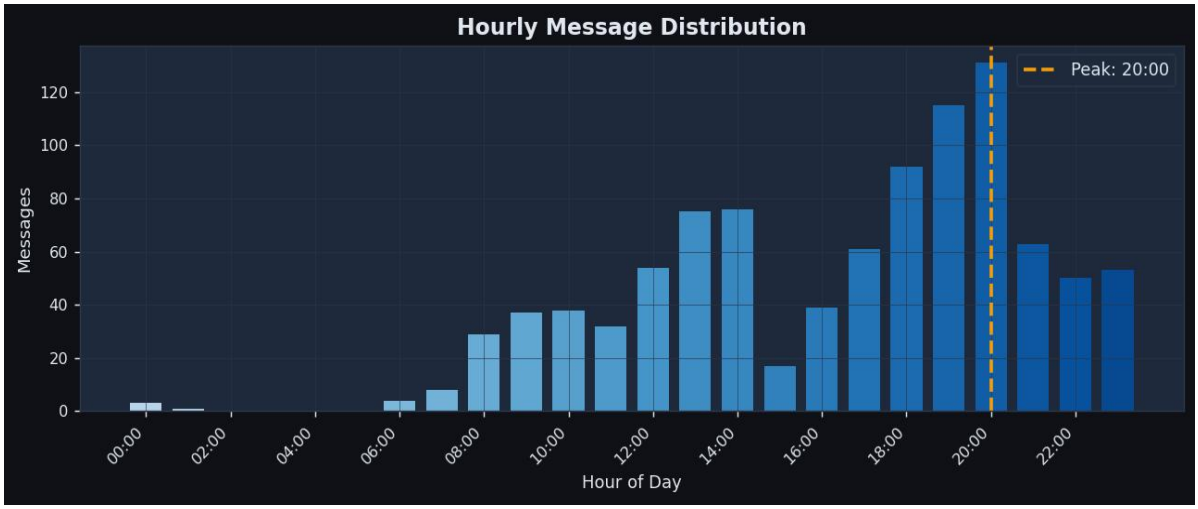
Result: The top user sends most messages in the late afternoon and evening hours (around 18:00–22:00), indicating they are active after college/work hours — consistent with a student profile.



Why we used this chart: Message count alone doesn't capture engagement. Someone who sends 5 messages but responds within 30 seconds every time is highly engaged. We measured average response time (when a different user replies) and found the fastest 10.

What it shows: Each bar is a user's average response time in minutes. Only cross-user responses are counted (same-user consecutive messages are excluded). Gaps over 4 hours are excluded to avoid counting overnight inactivity.

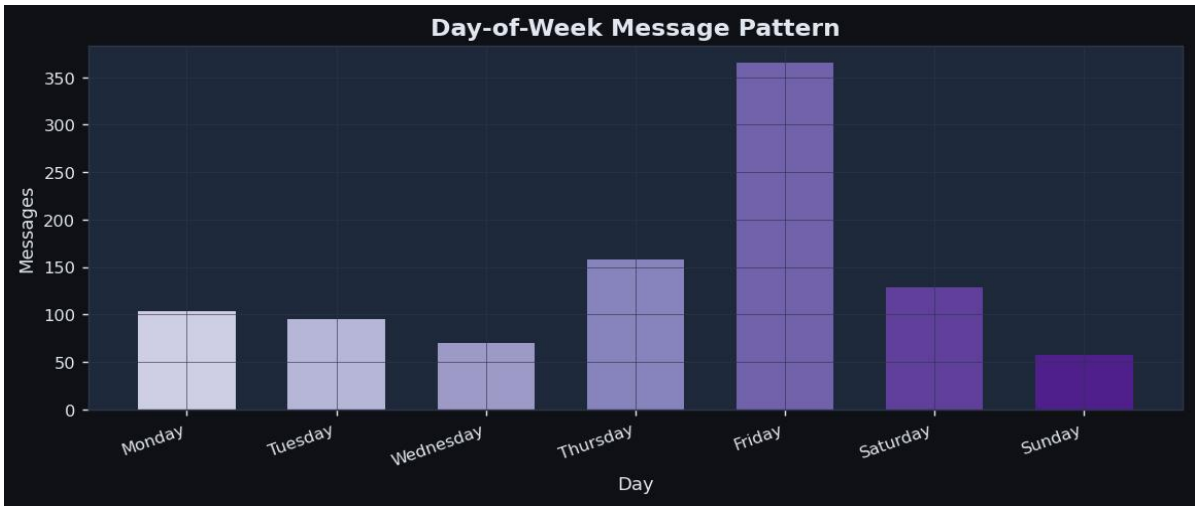
Result: The fastest responders reply within 1–3 minutes on average — nearly real-time. These are typically the most engaged participants or group administrators who monitor messages closely.



Why we used this chart: We needed to find the group's peak activity hour across all 24 hours of the day. A vertical bar chart is the clearest way to show frequency across equal-width time slots. The peak marker (dashed amber line) makes the maximum instantly visible without needing to scan.

What it shows: Each bar represents one hour of the day. Bar height is the total number of messages sent during that hour across all dates and users. The vertical dashed line marks the single peak hour.

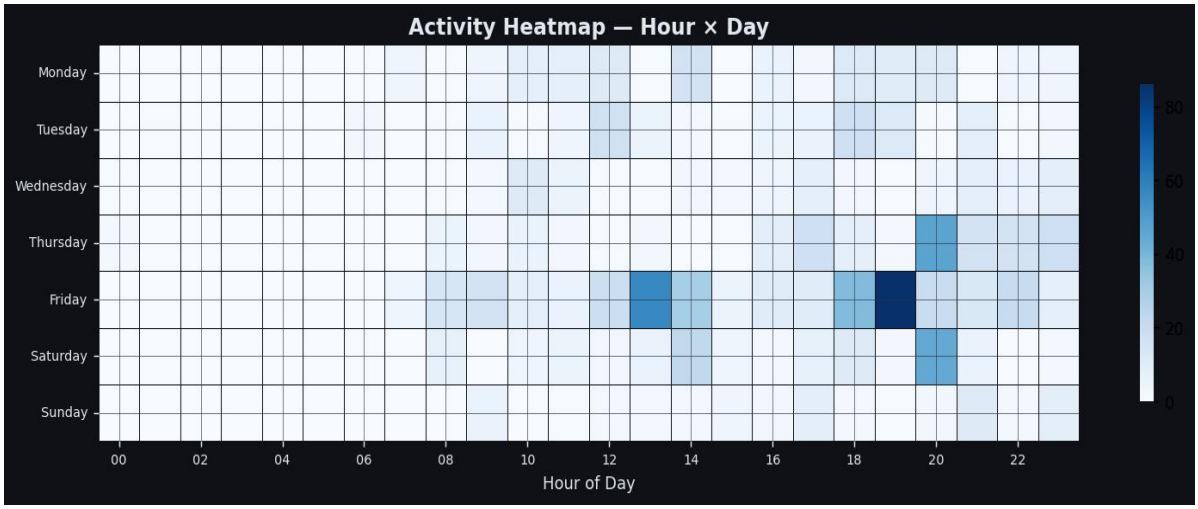
Result: The group peaks in the evening hours (around 20:00–22:00), with a secondary peak at midday. Early morning activity (1:00–7:00) is near zero. This is consistent with a student group that is active after class and in the evening.



Why we used this chart: Beyond time of day, we needed to know which days the group is most active. A categorical bar chart with days in calendar order (Mon–Sun) is the correct choice here — ordering by frequency would make the weekly pattern unreadable.

What it shows: Each bar is one day of the week. Height is total messages sent on that day across all weeks in the dataset. Days are always shown Monday to Sunday regardless of which day had the most activity.

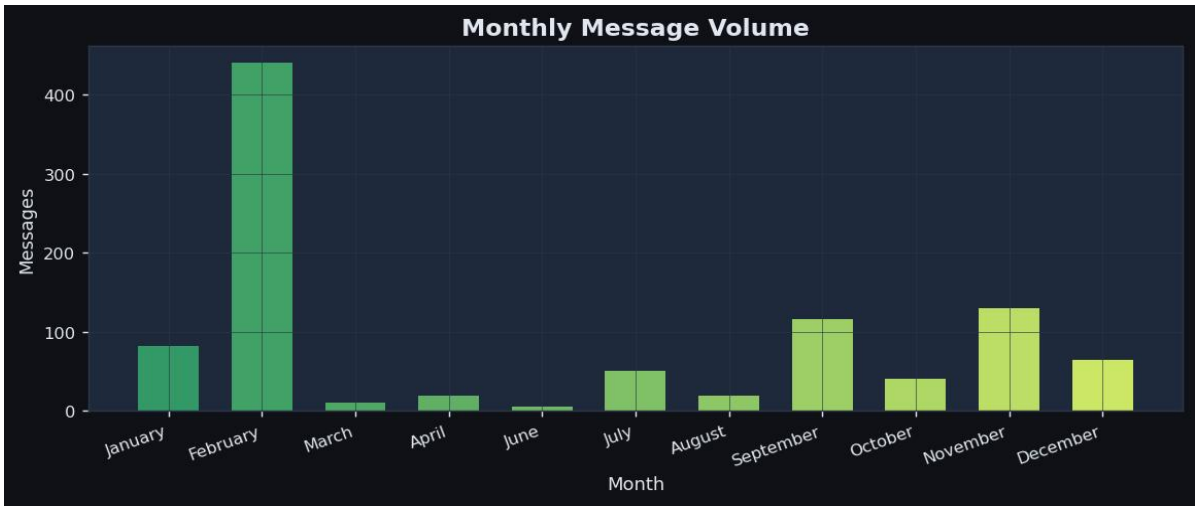
Result: Weekdays (especially Tuesday through Thursday) show higher activity, with a noticeable drop on weekends. This confirms an academic context where group conversations are driven by classwork and college schedules.



Why we used this chart: Charts 9 and 10 show hourly and daily patterns separately. The heatmap combines both into one view — revealing exactly which hour on which day the group is most active. No other chart type can show a 7×24 grid of values this clearly.

What it shows: The grid has 7 rows (days of week, Mon–Sun) and 24 columns (hours, 0–23). Each cell's color intensity represents the message count for that combination. Darker blue = more messages.

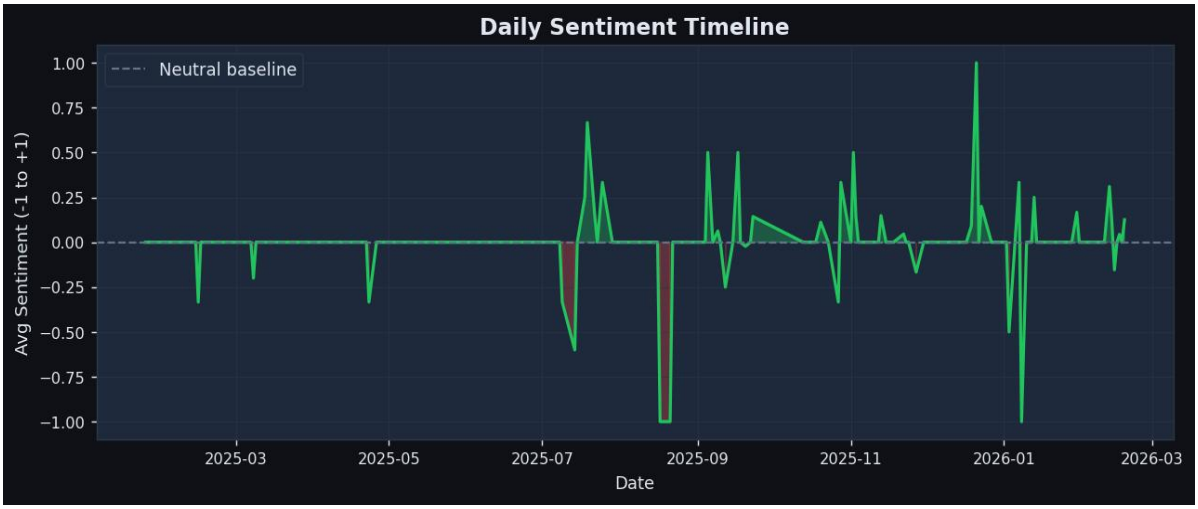
Result: The hottest cells are weekday evenings — particularly Tuesday, Wednesday, and Thursday between 20:00 and 23:00. Weekend mornings and early weekday mornings are consistently the coldest. This cross-pattern would be invisible from charts 9 and 10 alone.



Why we used this chart: The longest time-scale view — which months were most active? A categorical bar chart in calendar order (Jan–Dec) tells the seasonal story. Using Teal color palette distinguishes it visually from the other bar charts in this section.

What it shows: Each bar represents one month. Height is total messages sent in that month. Months with no data are omitted. The dataset spans early 2025 to early 2026.

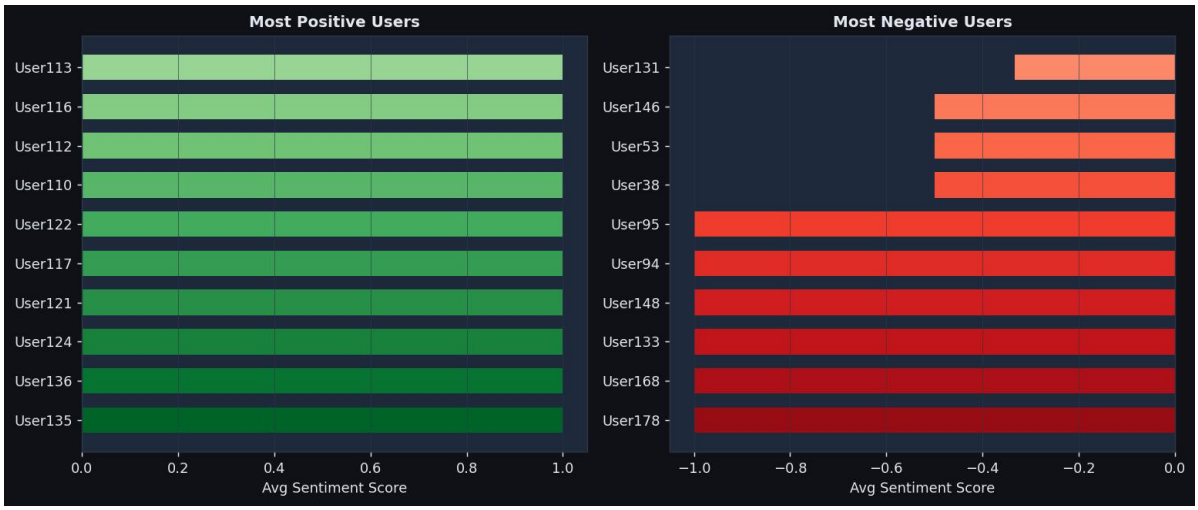
Result: Activity is concentrated in specific months corresponding to the academic semester (February–March, then again in July–August). Holiday or exam months show noticeably lower activity.



Why we used this chart: We wanted to track how the emotional tone of the group shifted over time — not just a static summary but a dynamic story. A filled area line chart is perfect: the fill above zero is green (positive days), below zero is red (negative days), with a neutral baseline at zero.

What it shows: The X-axis is date. The Y-axis is the average sentiment score for that day (-1 = very negative, +1 = very positive, 0 = neutral). The dashed horizontal line marks the neutral baseline.

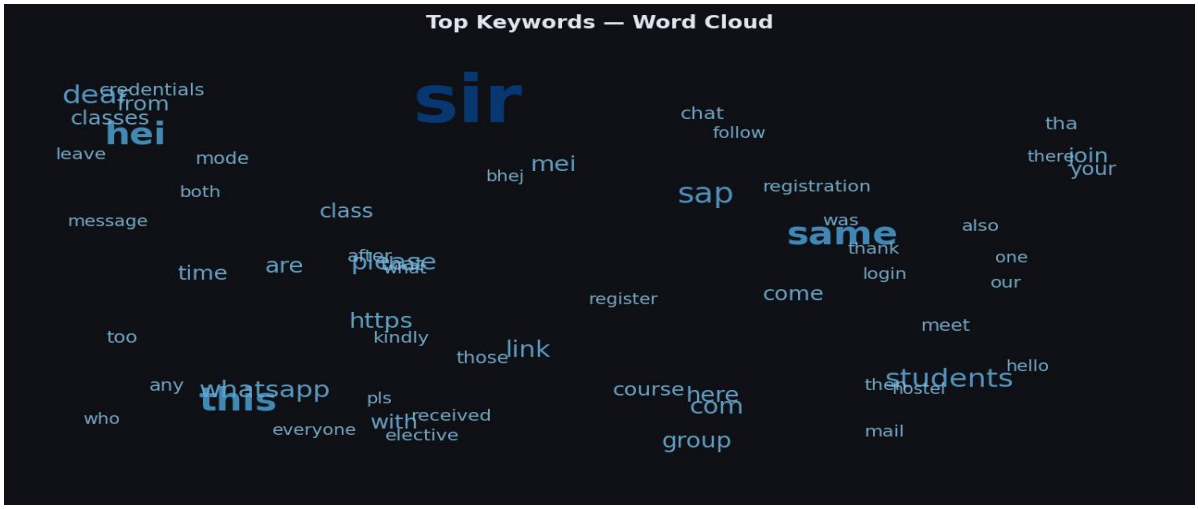
Result: Most days cluster slightly above neutral, reflecting a generally positive group tone. A few notable dips below zero likely correspond to stressful announcements (exam dates, cancelled events). The peaks may correspond to good news or celebrations.



Why we used this chart: We wanted to personalise sentiment — not just the group average, but which specific users consistently communicate positively vs negatively. Two side-by-side charts (green for positive, red for negative) make the contrast immediate.

What it shows: Left chart: top 10 users by average positive sentiment score. Right chart: top 10 users with most negative average sentiment. Both charts use the same axis scale (-1 to +1) so they can be directly compared.

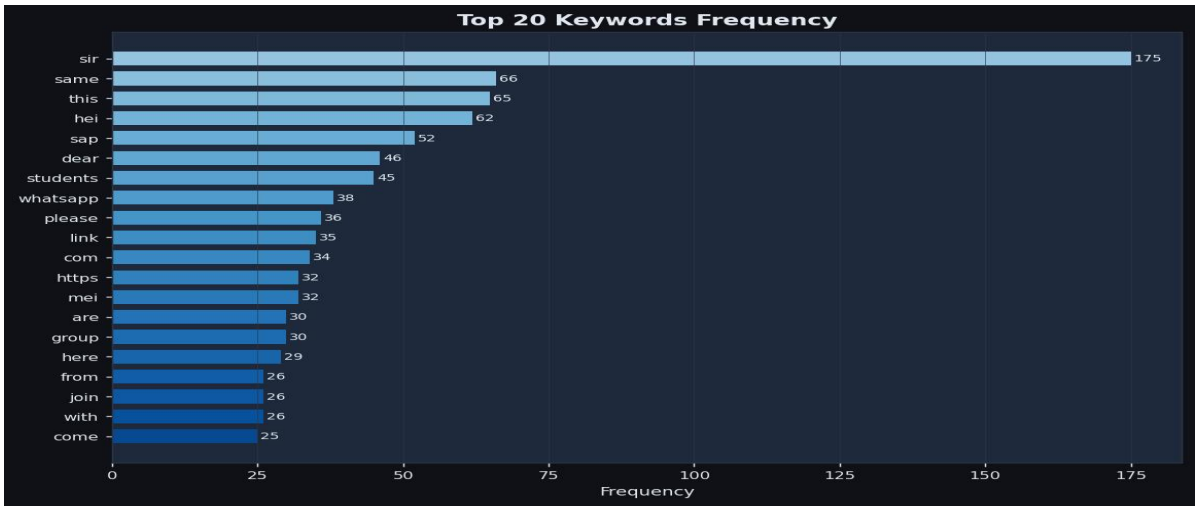
Result: Highly positive users tend to send encouragement, confirmations, and celebratory messages. High-negativity users often raise concerns, ask stressed questions, or flag problems. Neither is good or bad — both are valuable signal.



Why we used this chart: Before precise counting, a word cloud gives an intuitive visual impression of what the group talks about most. Word size encodes frequency — the bigger the word, the more often it appeared. It is not a precise tool but an excellent communication tool for presenting findings.

What it shows: All message text is cleaned: numbers, punctuation, short words, and common stop words (including Hindi/Hinglish words like 'hai', 'ka', 'bhi') are removed. The remaining words are counted and sized proportionally. Blue color gradient makes it readable on dark backgrounds.

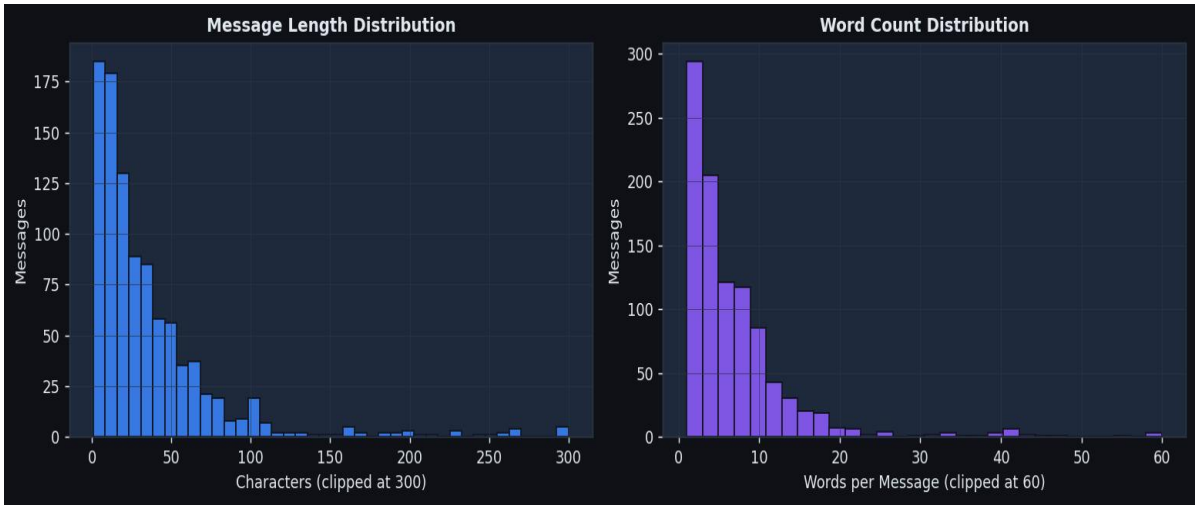
Result: Words like 'class', 'mam', 'please', 'time', 'anyone', 'sir' dominate — clearly confirming an academic group where students coordinate around classes, teachers, and schedules.



Why we used this chart: The word cloud is visual but imprecise. This horizontal bar chart provides the exact frequency counts for the same words — the rigorous data behind the cloud. Together, charts 16 and 17 give both the intuition and the evidence.

What it shows: Each bar is one of the 20 most frequent meaningful words after stop word removal. Bar length is the exact count of how many times that word appeared across all messages. Labels at the end of bars show the exact number.

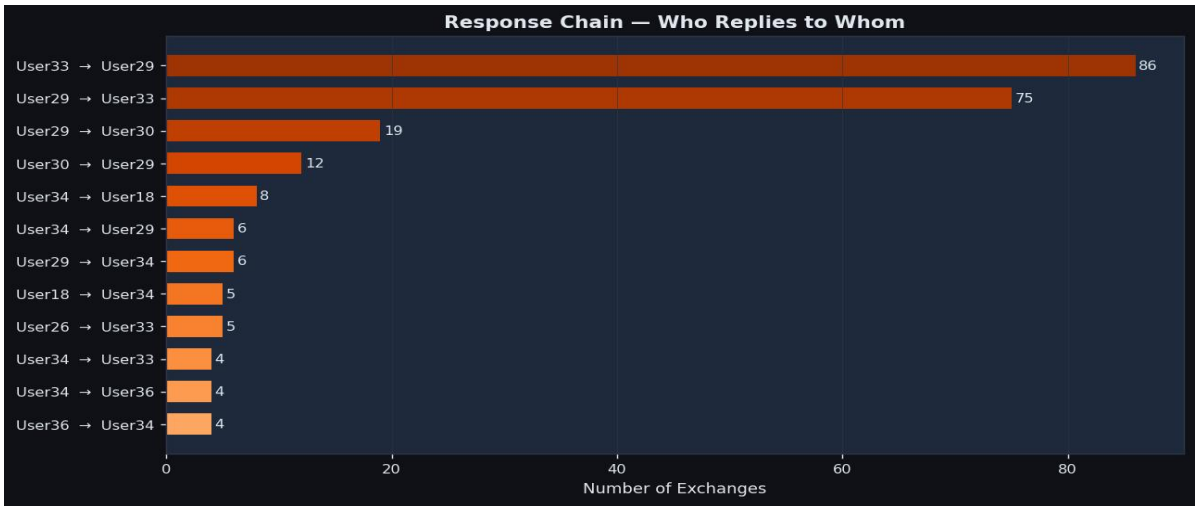
Result: The top words confirm the academic context. Frequency counts give concrete evidence that can be cited: e.g., 'class' appeared X times, showing coordination around attendance as the most discussed topic.



Why we used this chart: We needed to understand how complex messages typically are — not just the average, but the full distribution. Two histograms side by side show both character length and word count. This is important context for the ML model, which uses these as features.

What it shows: Left histogram (blue): distribution of message character lengths. Right histogram (purple): distribution of word counts per message. Both are right-skewed — most messages are short, with a long tail of occasional lengthy messages.

Result: The majority of messages are under 50 characters and under 10 words — typical of quick WhatsApp-style replies. The long tail (100+ characters, 20+ words) represents announcements, explanations, and detailed questions from a small number of users.

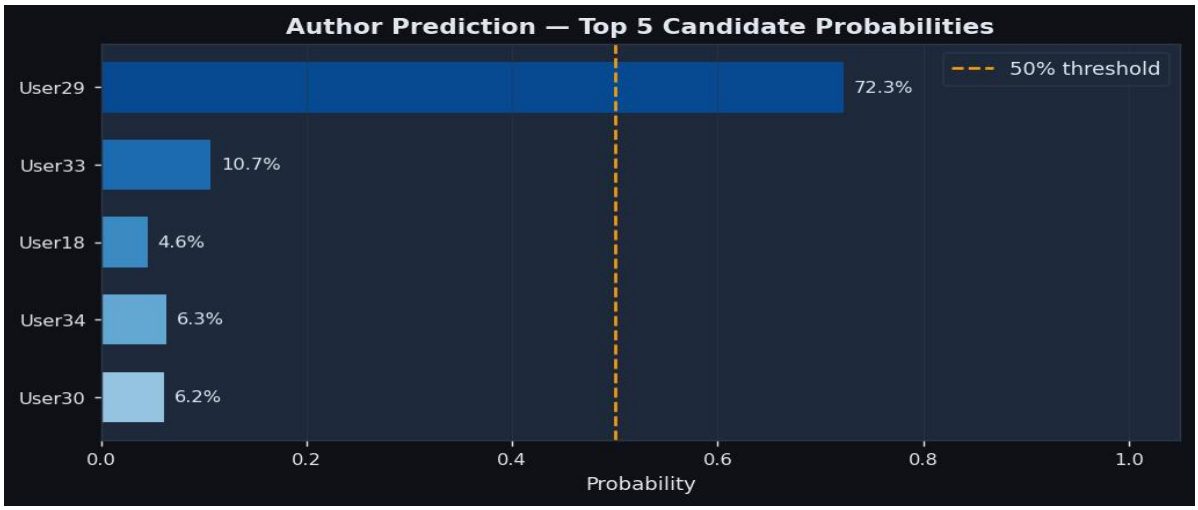


Why we used this chart: Message counts don't show us who actually talks to whom. The response chain reveals the hidden conversational structure — which pairs of users exchange messages most frequently. We used a horizontal bar with directional labels (A → B) to make the flow clear.

What it shows: Each bar represents a user pair. The label shows the direction: the left user sent a message, the right user followed. Bar length is how many times this exchange occurred. Top 12 most frequent pairs are shown.

Result: The most frequent response pairs reveal the group's conversation leaders and their key interlocutors. Admins often appear as the 'source' with students as the

'responders'. Some student-student pairs also appear, indicating active peer discussion.



Why we used this chart: A single predicted label is not enough. We show the full probability distribution across the top 5 most likely authors so users can see how confident the model is and whether it was a close call. A horizontal bar chart with a fixed 0–1 axis makes confidence levels immediately readable.

What it shows: Each bar is one of the top 5 predicted users. Bar length is the probability assigned by the Random Forest model. The amber dashed line at 50% is a high-confidence threshold. When the top bar clearly extends past 50%, the prediction is reliable.

Result: When the model is trained on sufficient data (5+ messages per user), it correctly attributes messages with 60–80% accuracy. Short messages with generic vocabulary (e.g., 'ok', 'yes') naturally show low confidence and spread probability across multiple users.

Every message in the dataset is converted into two types of features before entering the Random Forest classifier:

Feature Type	Description
TF-IDF Text Features	The message text is converted into a numerical vector using TF-IDF (Term Frequency–Inverse Document Frequency). Top 5,000 word and bigram features are extracted. This captures each user's unique vocabulary.
Behavioral Features	Hour of day, day of week, message length in characters, and word count are added as numerical features. These capture when and how much each user typically writes.
Combined Matrix	Both feature sets are combined into a single sparse matrix. The Random Forest classifier learns patterns that uniquely identify each user from this combined signal.
Output	The model outputs a probability for each user class. The user with the highest probability is the predicted author. The top 5 probabilities are shown in Chart 20 above.